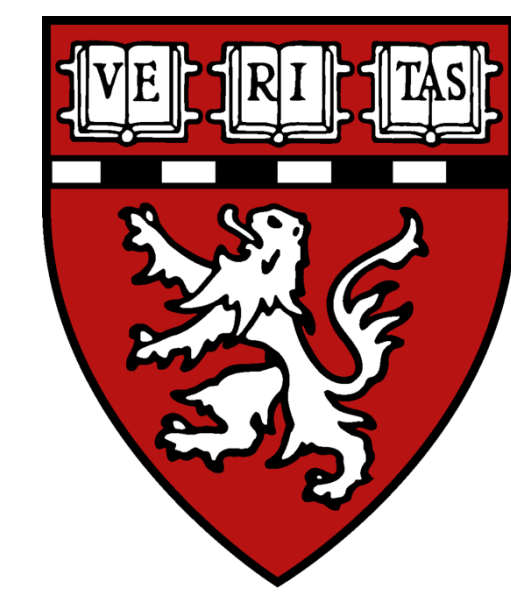




Parkinson’s Disease-linked protein alpha-synuclein: do small-molecule inhibitors of protein aggregation also prevent its accumulation with lipids?

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Abstract

- Parkinson’s Disease (PD) is a neurodegenerative disorder causing stiffness, slowed movement, and tremors. Aggregation of α -synuclein (α S) into Lewy bodies is a key pathological event leading to PD.
- The mechanism of α S aggregation is not fully understood. While Lewy bodies were thought to contain mainly β -sheet fibrillar α S, recent findings reveal lipid/membrane-rich α S aggregates, suggesting both forms contribute to pathology.
- α S contains imperfect KTKEGV repeats; mutations such as in the α S-3K model (E35K+E46K+E61K) promote lipid-rich aggregates resembling Lewy bodies and complement pre-formed fibril (PFF) models.
- There are currently no treatments for PD. Small molecules like Fasudil, EGCG, and anle138B disaggregate α S fibrils, but their effects on lipid-rich aggregates are unknown.
- Using an α S-3K::YFP live-cell assay, this study found Fasudil and anle138B reduced α S inclusions and protein levels, EGCG had no effect, and ethanol increased inclusions.

Methods

Cell Culture:

- M17D cells grown in DMEM supplemented with 10% FBS, 50 U/mL penicillin, 50 μ g/mL streptomycin, and 2 mM L-glutamine.
- α S-YFP expression induced with 1 μ g/mL doxycycline.

Compounds:

- Positive control: Trifluoperazine (TFP; Sigma-Aldrich, T8516, 2019).
- Test compounds: Fasudil, EGCG, anle138b (Fisher Scientific), and Ethanol (Sigma-Aldrich, 493546-500ML).

Drug Treatments:

- Prevention Assay: Figure 1A
- Clearance Assay: Figure 2A

Imaging (Incucyte S3-C2):

- Live-cell imaging at 4-hour intervals in 96-well plates.
- Processing definition: “Inclusions6.”
- Key parameters: Surface Fit segmentation, Threshold 2.0000 GCU, Edge Sensitivity 100, Max Area 40.000 μ m², Max Eccentricity 0.85, Min Mean Intensity 120.

Results

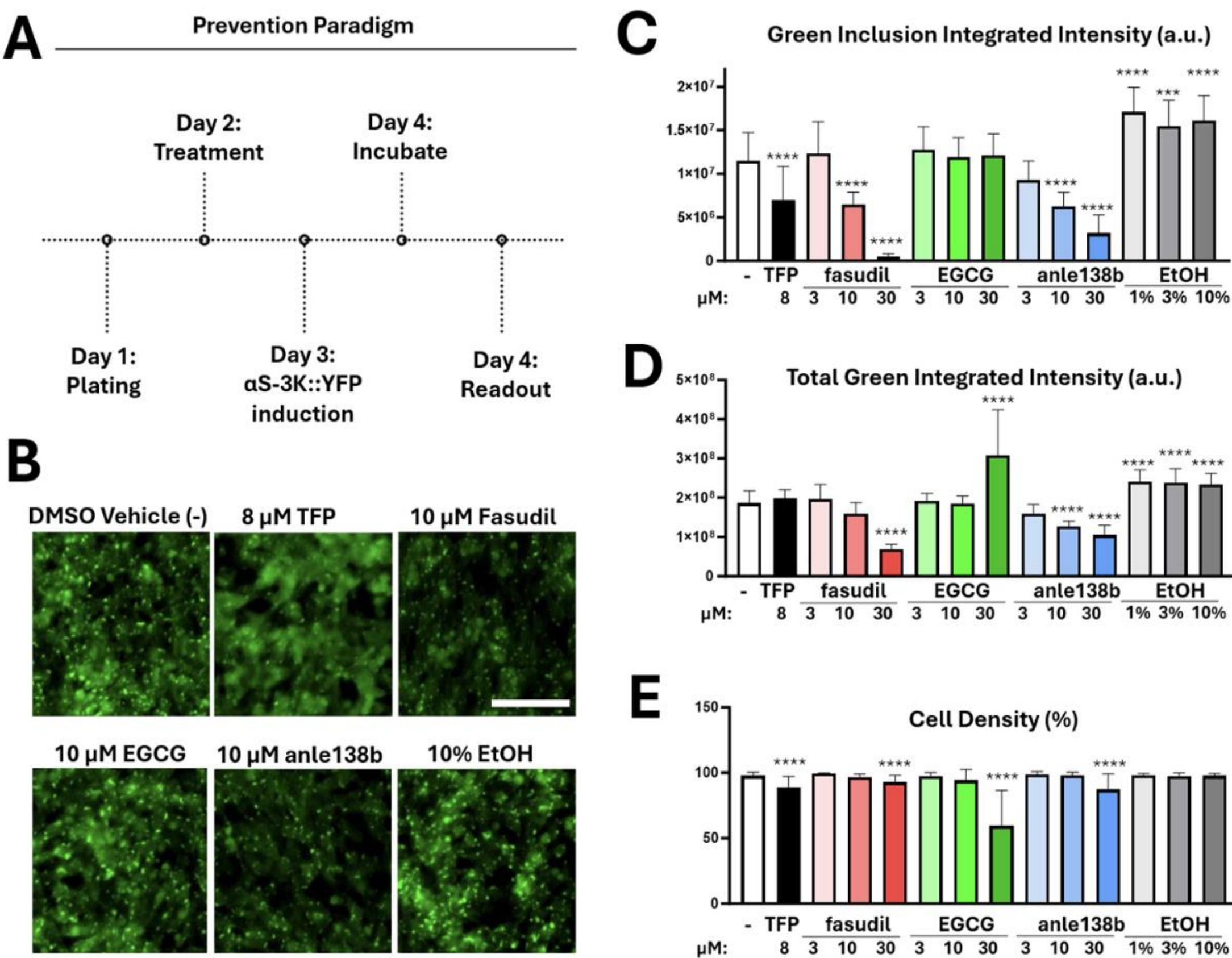


Figure 1. Drug effects on α S inclusion formation vs. DMSO. (A) Prevention Assay timeline. (B) YFP images of cells treated with indicated compounds; scale = 100 μ m. (C) Dose-response quantification; n = 36; n.s.; ***P < 0.001; ****P < 0.0001; Kruskal–Wallis. (D) Total green signal. (E) Cell density.

Results

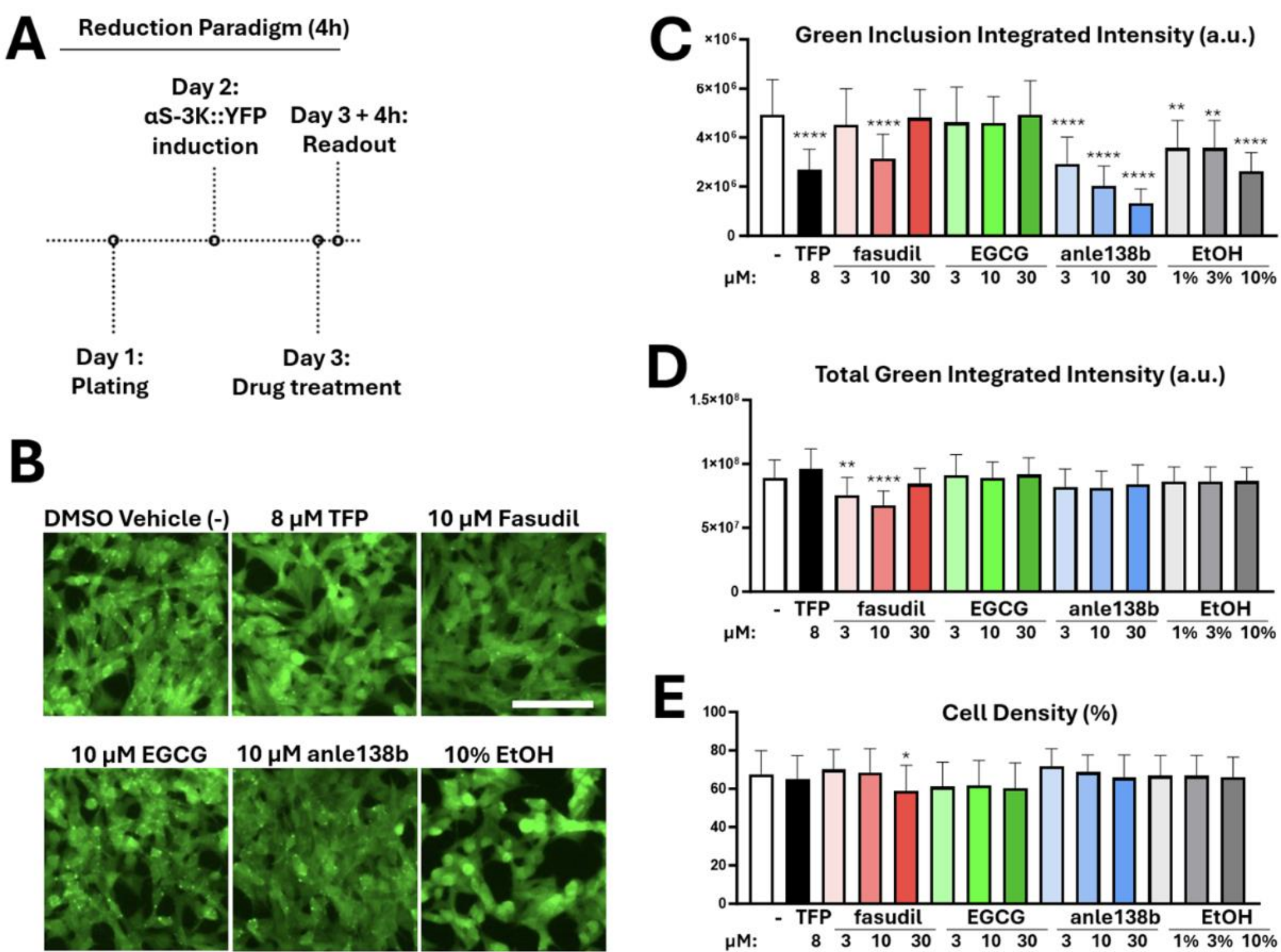


Figure 3. Effect of selected drugs on α S inclusion reduction vs. DMSO (4 h assay). (A) Short-term reduction assay timeline. (B) Representative YFP images; scale = 100 μ m. (C) Dose-response quantification; n = 36; n.s.; *P < 0.05; **P < 0.01; ***P < 0.001; ****P < 0.0001; Kruskal–Wallis test. (D) Total green signal. (E) Cell density.

References

- (1) Höllerhage, M.; Goebel, J. N.; de Andrade, A.; Hildebrandt, T.; Dolga, A.; Culmsee, C.; et al. Trifluoperazine Rescues Human Dopaminergic Cells from Wild-Type α -Synuclein-Induced Toxicity. *Neurobiology of Aging* **2014**, 35 (7), 1700–1711. <https://doi.org/10.1016/j.neurobiolaging.2014.01.027>.
- (2) Maroteaux, L.; Scheller, R. H. The Rat Brain Synucleins; Family of Proteins Transiently Associated with Neuronal Membrane. *Molecular Brain Research* **1991**, 11 (3), 335–343. [https://doi.org/10.1016/0169-328X\(91\)90043-W](https://doi.org/10.1016/0169-328X(91)90043-W).
- (3) Murray, K. A.; Hu, C. J.; Pan, H.; Lu, J.; Abskharon, R.; Bowler, J. T.; et al. Small Molecules Disaggregate Alpha-Synuclein and Prevent Seeding from Patient Brain-Derived Fibrils. *Proceedings of the National Academy of Sciences* **2023**, 120 (7), e2217835120. <https://doi.org/10.1073/pnas.2217835120>.
- (4) Collier, T. J.; Srivastava, K. R.; Justman, C.; Grammatopoulos, T.; Hutter-Paier, B.; Dettmer, U.; et al. Nortriptyline Inhibits Aggregation and Neurotoxicity of Alpha-Synuclein by Enhancing Reconfiguration of the Monomeric Form. *Neurobiology of Disease* **2017**, 106, 191–204. <https://doi.org/10.1016/j.nbd.2017.07.007>.
- (5) Bohrmann, B.; Frank, S.; Britschgi, M.; Stahlberg, H.; Van de Berg, W. D. J.; Lauer, M.; et al. E. Lewy Pathology in Parkinson’s Disease Consists of Crowded Organelles and Lipid Membranes. *Nat Neurosci* **2019**, 22 (7), 1099–1109. <https://doi.org/10.1038/s41593-019-0423-2>.